

CS6440: Introduction to Health Informatics Spr24

Practicum Sprint # 2 - Project Planning

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Abstract—We introduce *SkinWise*, a comprehensive web-based diagnostic tool designed to enhance the capabilities of healthcare providers, including hospitals and doctors, in the early detection and efficient management of skin cancer, with a particular focus on melanoma. SkinWise incorporates advanced machine learning techniques, leveraging the Skin Cancer ISIC dataset and Fast Healthcare Interoperability Resources (FHIR) to provide an innovative solution for the accurate classification and management of skin cancer.

PART 1 - PROJECT TOPIC

Background

Skin cancer is a major public health issue, with its incidence increasing globally. Among various types, melanoma is particularly notable for its severity and high mortality risk if not identified early (Khazaei et al., 2019). The standard approach to skin cancer diagnosis has traditionally relied on visual inspection and dermoscopic analysis of skin lesions (Ruffano et al., 2018). The accuracy of such diagnoses often varies, largely due to the subjective nature of the assessment, which is dependent on the individual healthcare professional's experience and expertise. This variability can lead to diagnostic inaccuracies.

The challenge of diagnosing skin cancer lies in the subtlety of its manifestations and the requirement for specialized knowledge to interpret dermoscopic images accurately (Loescher et al., 2013). Given these complexities, the proposed project, SkinWise, aims to leverage advancements in artificial intelligence (AI) and a comprehensive dataset from the Skin Cancer ISIC Dataset available at . By integrating these resources with Fast Healthcare Interoperability Resources (FHIR), SkinWise seeks to augment the current diagnostic processes and improve patient management strategies (Ruffano et al., 2018). This initiative offers a promising solution to the challenges of skin cancer diagnosis by enhancing the objectivity, accuracy, and efficiency of the process (Patel et al., 2023).

Justification

The deployment of an automated skin cancer detection system (Dildar et al., 2021), such as SkinWise, is critical in addressing the global disparities in access to specialized dermatological care. Conventional diagnostic methods' subjective nature makes it unreliable, emphasizing the need for a standardized and objective diagnostic tool (Naqvi et al., 2023). SkinWise addresses these challenges by providing a web-based diagnostic platform that employs deep learning algorithms trained on a vast dataset. This approach aims to standardize skin cancer diagnosis. By integrating with FHIR standards, SkinWise also ensures seamless interoperability within the healthcare system.

Solution

SkinWise proposes a state-of-the-art solution to the challenges of skin cancer diagnosis through the development of a web-based platform. This platform will utilize a deep learning model trained on the Skin Cancer ISIC Dataset to analyze images of skin lesions and accurately identify various skin conditions. The integration of FHIR standards (Mandel et al., 2016) will ensure that SkinWise is compatible with diverse electronic health record systems, promoting interoperability and efficient data exchange within the healthcare ecosystem. The platform will enable healthcare professionals to upload patient images and receive diagnostic suggestions in real-time.

PART 2 - TECHNICAL DESIGN

Tools/Technology

Below is a list of tools and technologies that will be used in our SkinWise project, along with brief descriptions of why they were chosen:

- **Python:** A versatile programming language selected for its extensive libraries and community support in backend development, data analysis, and machine learning classification model development.
- **PyTorch/TensorFlow/Keras:** These libraries are for developing and training machine learning models. PyTorch and TensorFlow provide comprehensive ecosystems for deep learning, while Keras simplifies neural network creation.
- **Flask:** A lightweight Python web application framework, chosen for its simplicity and flexibility in developing backend services.

- **HTML/CSS/JavaScript:** Core technologies for building the frontend, ensuring the application is responsive and interactive.
- **PostgreSQL:** An open-source relational database system, selected for its reliability and performance in managing application data.
- **Heroku:** A cloud platform as a service (PaaS) for easy deployment and scaling of web applications.
- **FHIR (Fast Healthcare Interoperability Resources) standards:** Adopted to ensure interoperability and secure communication of health information within the application.

Datasets and Data Sources

We will utilize the following datasets and data sources in this project:

- **FHIR (Fast Healthcare Interoperability Resources):** To ensure interoperability in healthcare applications.
- **Skin Cancer ISIC dataset from Kaggle:** A comprehensive dataset of skin cancer images used for training the machine learning model.
- **Synthetic test data generated from the Skin Cancer ISIC dataset:** For testing purposes, ensuring the model's robustness and accuracy.

Architecture Diagram

We provide a straightforward architecture diagram to illustrate the overall structure of our SkinWise solution. As shown in Figure 1, it includes the frontend, backend, the skin cancer classification ML model, database, and the integration of FHIR standards for data interoperability. We use Lucidchart to create our architecture diagram.

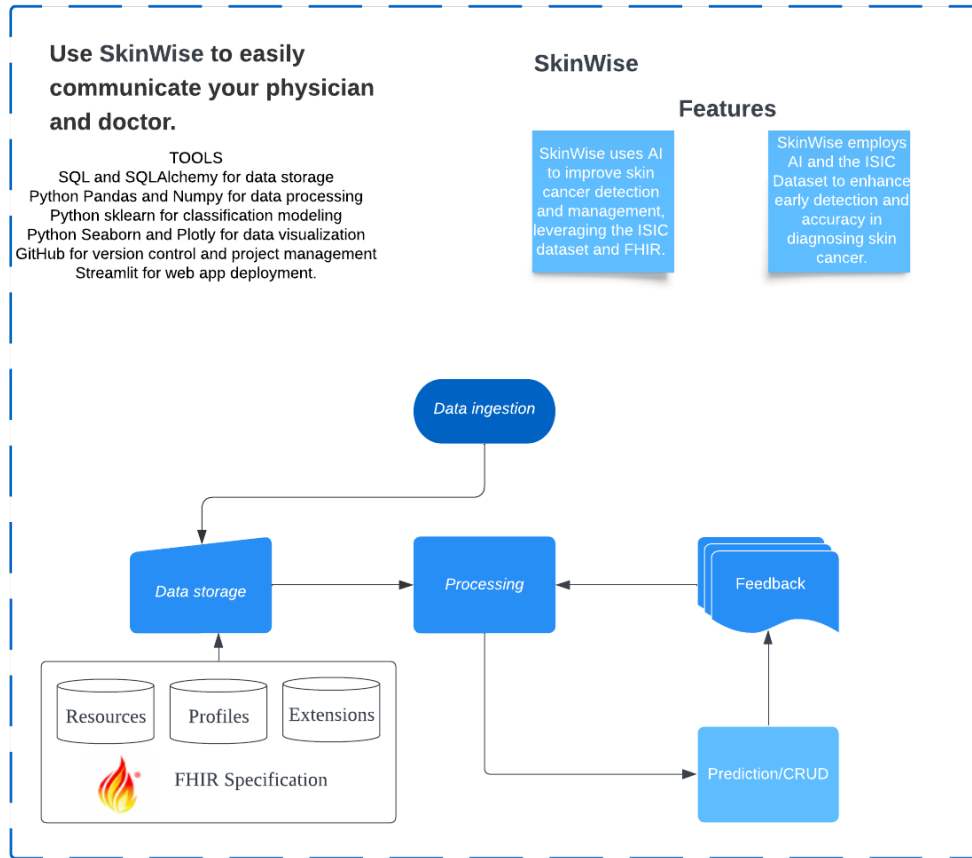


Figure 1—Architecture Diagram of SkinWise

Screen Mock-up(s)

Since SkinWise is a web-based diagnostic tool, we also provide screen mock-ups for showcasing the user interface (UI) and user experience (UX) design. We also use Lucidchart to design our mock-up illustrations. Figure 2 indicates our initial idea of the home page of SkinWise.

Specifically, we consider the following items in the home page design:

- Home Page: The landing page that introduces the application.
- Data Input/Upload Screen: Allows users to input or upload data (e.g., medical images).
- Results Display Screen: Displays the analysis results, incorporating FHIR standards for data presentation.

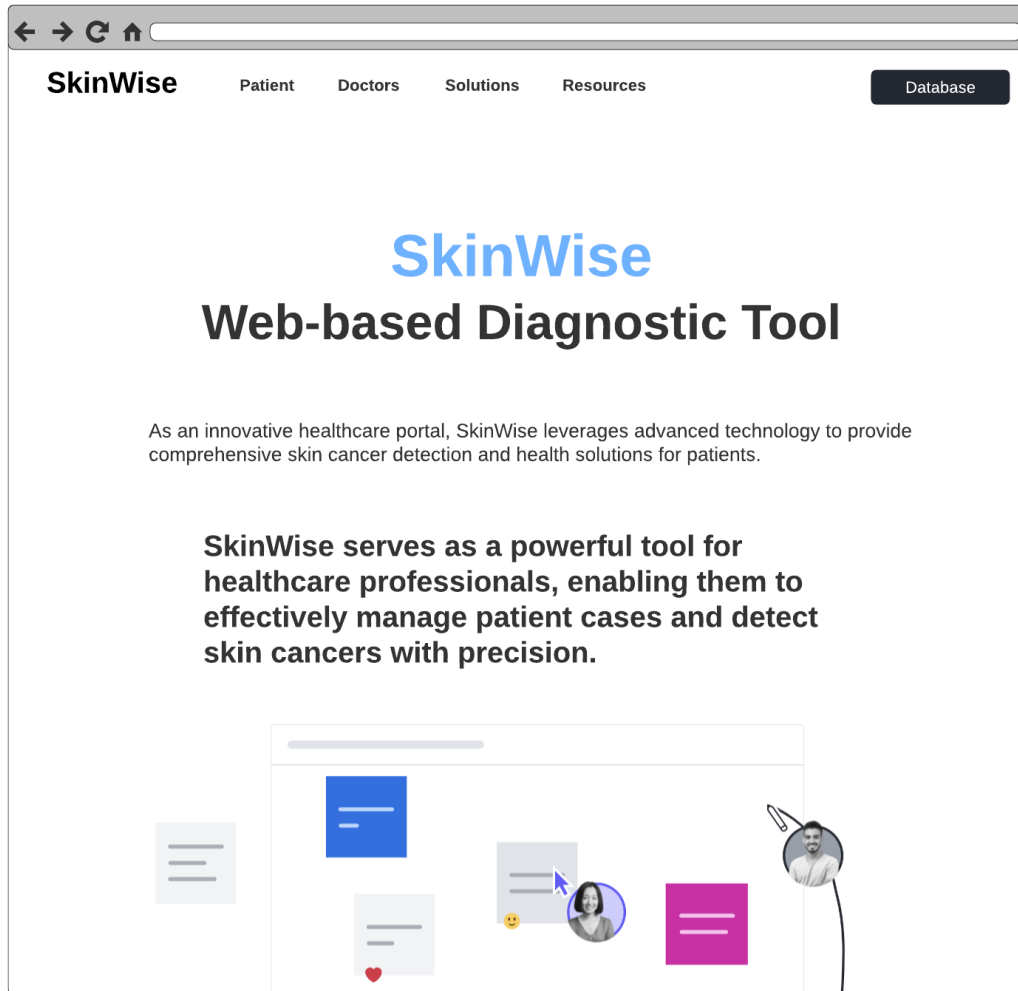


Figure 2—Screen Mock-up 1: homepage design of SkinWise

- User Dashboard: (If applicable) Shows historical data, analyses, and personalized information for registered users.

We also designed another mock-up for web elements as shown in Figure 3. Our design incorporates several modules, notably comprising a navigation bar, a patient information panel, skin images, machine learning model classification results, as well as summary and trend analysis charts.

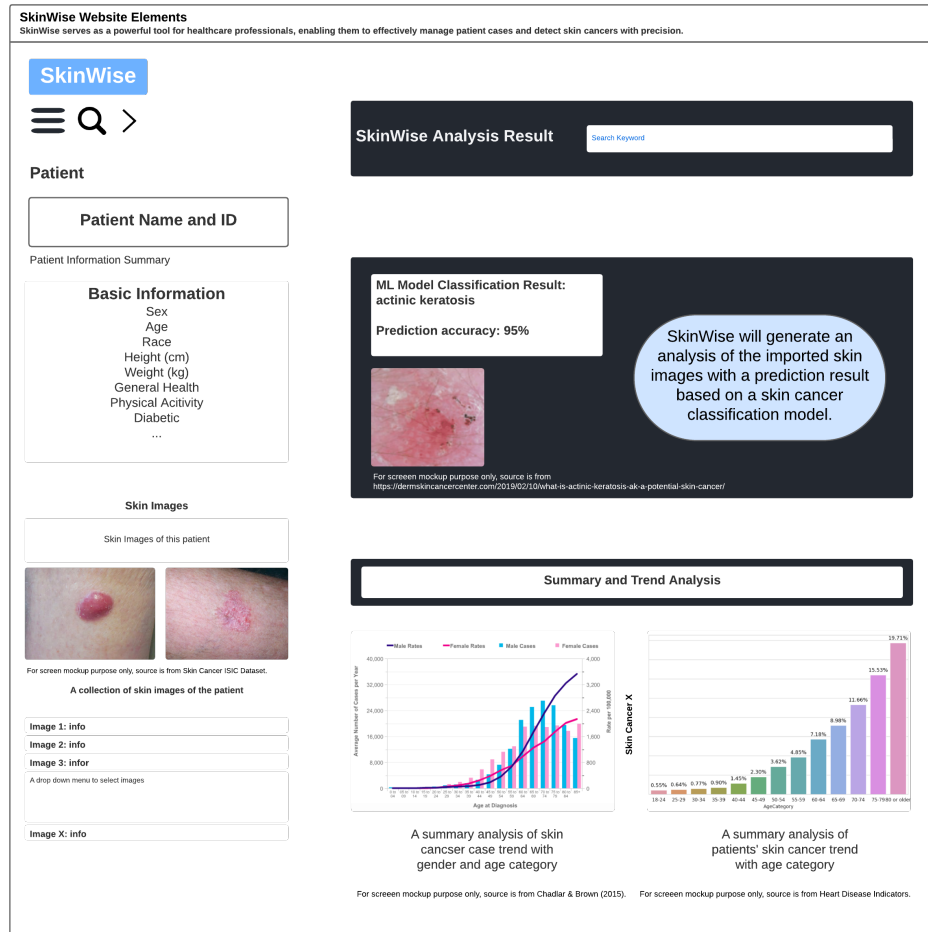


Figure 3—Screen Mock-up 2: elements in the information retrieval page of SkinWise

PART 3 - IMPLEMENTATION PLAN

In this project, we have delineated a set of high-level tasks that align with the course timeline, and needs and risks. These tasks have been structured to ensure alignment with the overall goals and to facilitate the successful completion of the project timeframe. Table 1 provides a comprehensive overview of our detailed implementation plan for the SkinWise project. We specify the project tasks, a timeline outlining the tentative date period for each task, and a thorough assessment of the needs and potential risks associated with each task. This detailed plan serves as a roadmap for our project team, guiding our efforts as we work towards the development and implementation of the SkinWise platform.

Project Timeline	Project Tasks	Needs	Risks
Jan 8 - Jan 21	Work to form our team.	Identify potential team members with similar project interests and diverse skills and expertise. Establish clear communication channels for team collaboration.	Difficulty in finding suitable team members with the close interests. Communication challenges within the team, e.g., time zones.
Jan 22 - Jan 28	Team formation and project kick off. Submit Practicum sprint # 1.	Formalize team structure and roles. Define project goals and deliverables.	Lack of clarity in project objectives leading to confusion among team members. Initial setbacks in team coordination and workflow.
Jan 29 - Feb 18	Brainstrom team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.	Research resources and literature relevant to our SkinWise project. Brainstorm the solution. Assign specific roles and responsibilities to team members.	Difficulty in finding relevant and reliable literature/resources. Difficulty in finding relevant health data sources. Potential conflicts arising from role assignments within the team.
Feb 19 - Feb 25	Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.	A project plan outlining the scope, objectives, and timeline of the SkinWise web application. Discussions and consultations with mentor to gather feedback. Explore various architecture design options for the SkinWise. Update our sprint # 2 with mentor's constructive feedback.	Difficulty in finding necessary data sources. Challenges in having interoperability. Shifting project focus leading to scope creep.
Feb 26 - Mar 3	Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.	Detailed project planning and design. Initiate data processing from the Skin Cancer ISIC Dataset. Set up the web app project environment.	Delays in project planning leading to a rushed implementation phase. Technical challenges in processing and handling the dataset. Technical challenges in setting the environment.
Mar 3 - Mar 10	Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.	Development of image classification model using deep neural networks. Initial setup of the web app codebase.	Complexity in implementing deep neural networks for image classification. Technical challenges in the web development with connecting with different datasets.
Mar 11 - Mar 17	Call for the patient resources from FHIR. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.	Access patient resources from FHIR for integration. Refine web app codebase. Conduct mentor check-in to ensure project alignment.	Delays or issues in obtaining necessary patient resources. Integrating data from FHIR sources can be complex. Relying on FHIR resources introduces dependencies.
Mar 18 - Mar 24	Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.	Continued project development including integration of FHIR data. Progress in web app development.	Difficulties in connecting FHIR patient data with the classification model. Challenges in the development of the web app functionality.
Mar 25 - Mar 31	Testing and validation of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.	Testing and validation of the web app. Final mentor check-in before project submission.	Inadequate testing leading to undiscovered bugs or issues. Insufficient time for addressing feedback from mentor check-in.
Apr 1 - Apr 7	Deployment of the SkinWise web application with CRUD functionality	Deployment of the SkinWise web application using CRUD operations. Examine the deployment online.	Deployment issues causing downtime or functionality failures. issues with data integrity, such as data loss or corruption, especially during data migration or database updates. Incomplete deployment leading to user dissatisfaction.
Apr 8 - Apr 14	Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.	Finalize project details and address any remaining issues. Prepare for project submission.	Online deployment may result in performance degradation, such as slow response times, increased latency, or system crashes, especially under heavy loads. Other challenges in the deployment such as configuration errors, network issues.
Apr 15 - Apr 21	Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.	Submission of the final project deliverables.	Technical difficulties or errors during submission. Incomplete or insufficient project documentations.

Table 1—Impementation plan: project tasks, project timeline, and identified needs/risks

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